



AI restores Rembrandt's painting

Rembrandt's *The Night Watch* restored by AI 300 years after it was trimmed to fit a smaller wall. <https://digit.in/jul21-13>



App turns users into spies

According to report, Premise pays users to complete tasks for gathering intelligence for the US military. <https://digit.in/jul21-14>

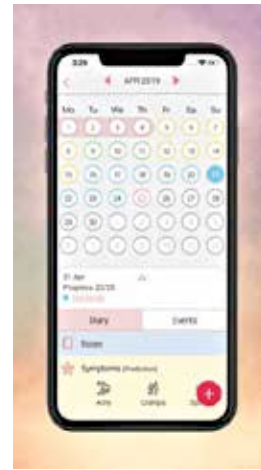
connected to, which discloses the general location of the user. Ikram explained that this information is collected and sometimes shared with third-party apps and companies for tracking and profiling purposes and more importantly, many of these apps do not explicitly take the prior permission of the users to disclose this data to third-party companies.

The researchers in this study also stated, "Our results show that the collection of personal user information is a pervasive practice in mHealth apps, and not always transparent and secure. Patients should be informed on the privacy practices of these apps and the associated privacy risks before installation and use." However, the study also noted that in comparison to other types of apps such as social media, shopping, and others, mobile health apps collect and share user data less often. Still, when the information is pertaining to health

all the health apps they studied, a whopping 28 per cent provided no valid policy text at all and about 25 per cent of data transmission violated what was stated in the privacy rules of the apps that had them. The study also found that 23 per cent of user-data transmissions occurred on insecure communication channels, which is a real cause for concern.

This is especially true since fitness apps, just like other apps, have suffered data breaches. For example, in 2018 there was a massive breach that hit MyFitnessPal and exposed the usernames, passwords, and email addresses of over 150 million users!

Users don't know what they're signing up for due to fitness apps'



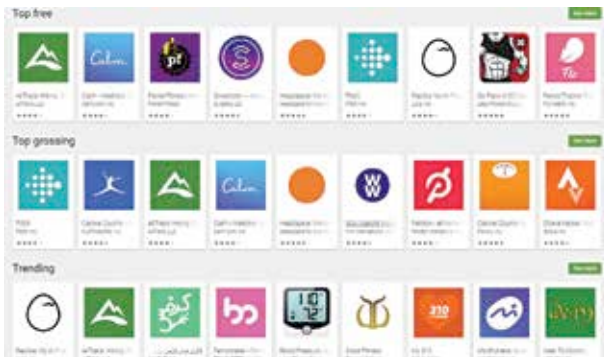
Maya, an ovulation and period tracking app

in fact, was sharing collected user data with several third parties, including the infamous Facebook. This is especially concerning considering the type of data Maya collects such as your contraception of choice, whether you're sexually active, and more.

To reduce the risks of health and medical data collected and distributed via these apps, a fundamental change in the business models

of these companies is absolutely necessary. Additionally, there needs to be some level of regulation and monitoring that keeps these apps in check when it comes to sharing sensitive information without the user's explicit consent. We also need to educate users of fitness wearables and apps about the risks of their data being leaked, in the meantime. At the moment, fitness apps and trackers collate information pertaining to our heart, lungs, sleep and overall fitness levels. But soon enough, say within the next 2-3 decades, it is not foolish to imagine that fitness data could even encompass critical information about our cognition.

It is irresponsible and inefficient to expect every individual to manage their risk personally, therefore the app developers and data collectors themselves must start conversing about ethics, data privacy, and responsible data management. The need for systematic reformation is high and this will possibly need to happen via legislation. One example is GINA (Genetic Information Nondiscrimination Act) which protects the constitutional right to privacy of American citizens' genetic information. Similar protections must be put in place for fitness, health and medical data collected by fitness trackers and health apps. 



Fitness tracking apps on the Google Play Store

and medical conditions, there's still a valid reason for concern.

Overall, 665 unique third-party entities were identified from the BMJ study sample, of which a small number of prominent third parties such as Google and Facebook were responsible for the most data collection.

LACK OF PRIVACY POLICIES

While collection and sharing of user data by health apps is a slightly routine affair, the transparency of these actions is questionable. The researchers of the BMJ (British Medical Journal) study found that of

information with our sponsors, and/or business partners". This is not a hypothetical privacy policy but an actual one found on the ovulation tracker app, Maya, which has over 8 million users worldwide! The privacy policy isn't explicit and doesn't give the user sufficient information on its practices to enable them to give informed consent.

Maya's vague privacy policy was exposed by a report by Privacy International, a UK-based registered charity that defends and promotes the right to privacy across the world. They also found that Maya,